25/09/26

Stratify

Cluster: A collection of data
↳ Similar to another within the same cluster
↳ Dissimilar to the subject

Characteristics of good clustering methods

Examples
marketing, Land use, Insurance, city-planning.

Two types of measures
1) Similarity → qualitative [0,1]
2) Dissimilarity [0, upper limit value]

Proximity refers to similarity or dissimilarity

Types of Variables
↳ Interval Scale variable
2) Binary
↳ ordinal, cardinal.

classification
- Hard
- Soft

Hierarchical
non-hierarchical
} clustering algorithms.

06/10/26

Cluster analysis

It is a family of method or techniques that group objects so that the objects in the same group/cluster are more similar to each other than to objects in other groups.

workflow :-

prepare data → measure proximity → choose method → form clusters →

choose number → Interpret.

Nominal – Matching (same / diff)

Binary → SMC, Jaccard

Ordinal → Ranks mapped to [0, 1]

Interval → Euclidean, manhattan

Ratio → Euclidean manhattan

} distance (or) proximity

Similarity s

How alike two objects are

Higher = more alike.

often in [0, 1]

Dissimilarity d

How two diff objects are.

Lower = more alike

minimum is often 0

Distance (metric)

1) $d(p, q) \geq 0$ non-ve

2) $d(p, q) = 0 \Leftrightarrow p = q$ identity

3) $d(p, q) = d(q, p)$ symmetry

4) $d(p, r) \leq d(p, q) + d(q, r)$ triangle inequality

Euclidean distance:-

$$d(p,q) = \sqrt{\sum_{k=1}^{n} (p_k - q_k)^2}$$

Minkowski:-

$$d(p,q) = \left[\sum_{k=1}^{n} |p_k - q_k|^r \right]^{1/r}$$

Mahalanobis distance:-

$$d_M(p,q) = \sqrt{(p-q)^T \Sigma^{-1} (p-q)}$$



Binary variable

Jaccard coeff

JMC:-

$$JMC = \frac{a+d}{a+b+c+d}$$

$$J = \frac{a}{a+b+c}$$

Count those is and those Ignores 0's focus on those S.

a = both 1

b = i=1 j=0 | c = i=0 j=1

d : both 0

08/09/26

Hierarchial step by step clustering

Agglomerative → each at every step as a single clus

Divisive → opp all the objects are on one cluster

Non Hierarchial

tree is at off position

Similarities

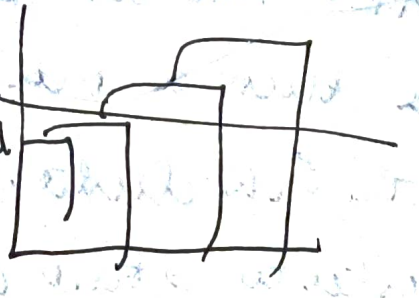
* Jaccard coeff
* matching coeff.

	Distance	Similarity matrix
meaning	How far apart	How alike.
Diagonal	0	1
Closest pair	smallest off-diagonal Value.	Largest off-diagonal Value.
typical chang	0 to ∞	0 to 1 (cosine (-1 to 1))
Symmetrical	Yes	Yes

Role:

For distance, small = closse, For similarity large = close.

Dendogram:



Dendogram:-

It is a tree like diagram that shows how observations or clusters are merged (or split) at diff level the vertical axis represents the dissimilarity at which the merge occurs.

Agglomerative:-

$d = 1 - s$

→ single linkage, complete, avg linkage method

* Agglomerative (bottom up)

Start: Every object is its own cluster

Repeat: merge the two closest cluster find one big cluster.

Divisive (top down)

*start: All objects in one cluster

repeat: split

Eg: small.

Single linkage cluster is based on min values of the distance matrix first the min value in the distance matrix is found if d[i,k] is min then ith and kth are merged to get them to form a cluster then a new distance matrix is formed with (i,k) cluster. the distance from (i,k) merge cluster to other are found then the

min of this new distance matrix is found & the process is repeated till all the objects all made to form one or more clusters.

1) Single linkage method:-

$d(i,k)s = \min\{dis, dks\}$.

2) Complete linkage method:-

$d(i,k)s = \max\{dis, dks\}$

3) Avg linkage method:

$d(i,k)s = \frac{dis + dks}{2}$.

Ward's methods
or
centroid method

The following is the distance matrix of stock prices of 6 companies

	1	2	3	4	5	6
1	0	0.398	0.362	0.952	0.941	0.855
2	0.398	0	0.421	0.978	0.965	0.944
3	0.362	0.421	0	0.994	0.994	0.999
4	0.952	0.978	0.994	0	0.329	0.392
5	0.941	0.965	0.994	0.329	0	0.352
6	0.855	0.944	0.999	0.392	0.352	0

Single Linkage method:

min & dis. desg.

D =	1	2	3	4	5
1	0	[2]	6	10	9
2	2	0	5	9	8
3	6	5	0	4	5
4	10	9	4	0	3
5	9	8	5	3	0

	(1,2)	3	4	5
(1,2)	0	5	9	8
3	5	0	4	5
4	9	4	0	3
5	8	5	3	0

	(1,2)	3	(4,5)
(1,2)	0	5	8
3	5	0	[4]
(4,5)	8	4	0

	(1,2)	[(4,5) 3]
(1,2)	0	5
[(4,5),3]	5	0

complete linkage method:

$D =$

	1	2	3	4	5
1	0	2	6	10	9
2	2	0	5	9	8
3	6	5	0	4	5
4	10	9	4	0	3
5	9	8	5	3	0

	(1,2)	3	4	5
(1,2)	0	6	10	9
3	6	0	4	5
4	10	4	0	3
5	9	5	3	0

	(1,2)	3	(4,5)
(1,2)	0	6	10
3	6	0	5
(4,5)	10	5	0

	(1,2)	((4,5),3)
(1,2)	0	10
((4,5),3)	10	0

Avg Linkage method:-

	1	2	3	4	5
1	0	2	6	10	9
2	2	0	5	9	8
3	6	5	0	4	5
4	10	9	4	0	3
5	9	8	5	3	0

	(1,2)	3	4	5
(1,2)	0	5.5	9.5	8.5
3	5.5	0	4	5
4	9.5	4	0	3
5	8.5	5	3	0

	(1,2)	3	(4,5)
(1,2)	0	5.5	9
3	5.5	0	[4.5]
(4,5)	9	4.5	0

	(1,2)	((4,5),3)
(1,2)	0	7.88/7.25
((4,5),3)	7.88/7.25	0

2)

	1	2	3	4	5	6
1	0	0.898	0.862	0.952	0.941	0.855
2	0.898	0	0.421	0.978	0.965	0.944
3	0.862	0.421	0	0.994	0.994	0.999
4	0.952	0.978	0.994	0	[0.829]	0.892
5	0.941	0.965	0.994	0.829	0	0.852
6	0.855	0.944	0.999	0.892	0.852	0

Single Linkage method :-

	1	2	3	(4,5)	6
1	0	0.898	0.862	0.941	0.855
2	0.898	0	0.421	0.965	0.944
3	0.862	0.421	0	0.941	0.999
(4,5)	0.941	0.965	0.994	0	[0.852]
6	0.855	0.944	0.994	0	0.852

	1	2	3	(4,5),6)
1	0	0.898	0.362	0.855
2	0.898	0	0.421	0.944
3	0.362	0.421	0	0.994
(4,5),6	0.855	0.944	0.994	0

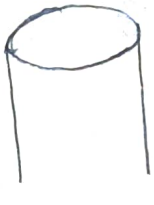
	(1,3)	2	(4,5),6
(1,3)	0	0.898	0.855
2	0.898	0	0.944
(4,5),6	0.855	0.944	0

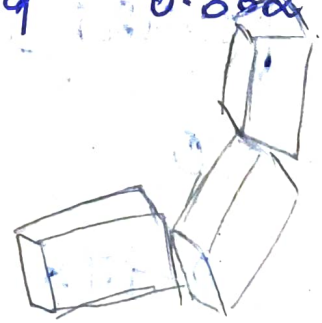
	(1,3),2	(4,5),6
(1,3),2	0	0.855
(4,5),6	0.855	0

complete - linkage method:-

	1	2	3	4	5	6
1	0	0.898	0.862	0.952	0.941	0.855
2	0.898	0	0.421	0.978	0.965	0.944
3	0.862	0.421	0	0.994	0.994	0.999
4	0.952	0.978	0.994	0	0.329	0.392
5	0.941	0.965	0.994	0.329	0	0.352
6	0.855	0.944	0.999	0.392	0.352	0

	1	2	3	(4,5)	6
1	0	0.898	0.862	0.941	0.855
2	0.898	0	0.421	0.965	0.944
3	0.862	0.421	0	0.994	0.999
(4,5)	0.941	0.965	0.994	0	0.352
6	0.855	0.944	0.999	0.352	0





	1	2	3	(4,5)	6
1	0	0.898	0.362	0.956	0.855
2	0.898	0	0.421	0.978	0.944
3	0.362	0.421	0	0.994	0.999
(4,5)	0.956	0.978	0.994	0	0.892
6	0.855	0.944	0.999	0.892	0

	(1,3)	2	(4,5)	6
(1,3)	0	0.421	0.994	0.999
2	0.421	0	0.978	0.944
(4,5)	0.994	0.978	0	0.892
6	0.999	0.944	0.892	0

	(1,3)	2	(4,5),6
(1,3)	0	0.421	0.999
2	0.421	0	0.978
(4,5),6	0.999	0.978	0

	(1,3),2	(4,5),6
(1,3),2	0	0.999
(4,5),6	0.999	0

Average linkage method :-

	1	2	3	4	5	6
1	0	0.398	0.362	0.952	0.941	0.855
2	0.398	0	0.421	0.978	0.965	0.944
3	0.362	0.421	0	0.994	0.994	0.999
4	0.952	0.978	0.994	0	0.329	0.392
5	0.941	0.965	0.994	0.329	0	0.352
6	0.855	0.944	0.999	0.392	0.352	0

	1	2	3	(4,5)	6
1	0	0.398	0.362	0.9465	0.855
2	0.398	0	0.421	0.9715	0.944
3	0.362	0.421	0	0.994	0.999
(4,5)	0.9465	0.9715	0.994	0	0.372
6	0.855	0.944	0.999	0.372	0

	(1,3)	2	(4,5)	6
(1,3)	0	0.4095	0.97025	0.927
2	0.4095	0	0.9715	0.944
(4,5)	0.97025	0.9715	0	0.872
6	0.927	0.944	0.872	0

	(1,3)	2	(4,5),6
(1,3)	0	0.4095	0.948625
2	0.4095	0	0.95775
(4,5),6	0.948625	0.95775	0

	(1,3),2	(4,5),6
(1,3),2	0	0.9531875
(4,5),6	0.9531875	0

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centroid method:-

First we calculate the centroid for all clusters. then define the distance b/w these centroids TO find the centroid it takes the avg value of each point within the cluster are determined

1) Treat each object or data point as a cluster
2) compute the centroids i.e., calculate the mean coordinate for each cluster
3) calculate the distance matrix i.e., measure the distance b/w the centroids of each pair of clusters
4) combine the two clusters that have the smallest distance b/w their centroids
5) calculate a new combined centroid using a weighted avg of the merged clusters
6) Repeat till you get one single cluster.

centroid method:-

$P_1 = 1, P_2 = 2, P_3 = 8, P_4 = 10$

Step1:- centrod: $C_1 = 1, C_2 = 2, C_3 = 8, C_4 = 10$

Step2:- $d(C_1, C_2) = |1-2| = 1$ $d(C_1, C_4) = |1-10| = 9$

$d(C_1, C_3) = |1-8| = 7$ $d(C_2, C_3) = 6$

$d(C_2, C_4) = 8$ $d(C_3, C_4) = 2$

Step3:- $d(C_1, C_2) = 1$ (smallest) merge (1,2)

$C_{12} = \frac{1+2}{2} = 1.5, C_3 = 8, C_4 = 10$

$d(c_1, c_3) = |1.5 - 8| = 6.5$

$d(c_1, c_4) = 1.5 - 10 = 8.5$

$d(c_3, c_4) = 2$ Smallest

$c_{34} = \frac{8+10}{2} = 9$

$d(c_1, c_{34}) = |1.5 - 9| = 7.5$

Ward's method:-

$$w(c) = \sum_{x \in c} \|x - \mu(c)\|^2$$

- ward merges the pair (c_a, c_b) with the smallest i'n

Common divisive splitting strategies

- DIANA
- Farthest-seed split
- Optimization-based split

Divisive hierarchical is a top down approach ML techniques its starts with all data points in a single large cluster by iteratively splitting into smaller distinct sub-cluster

Application:-

* Biological taxonomy
* customer segmentation in marketing
* product categorisation in ecommerce-e.g. online shopping website organised products using top down categorical structure

Monothetic Divisive clustering splits the clusters at each step based on single variable of feature the objects are divided into groups depending on whether they posses a specific value or threshold of that single attribute every member within a resulting sub population shares a exact explicit common property defined by the plot of splitting rules. The resulting dendogram acts like a decision tree.

Polythetic divisive clustering splits the cluster each step by evaluating all variables or features simultaneously using distance or dissimilarity measures. The algorithm calculates overall multidimensional distance.

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Non-Hierachial clustering analysis

Its an unsupervised ML techniques that divides a dataset into predefined no (K) of non overlapping well defined groups. clustering methods are iterative method

Procedure:-

1) We have to choose of clusters in advance (specify the values for 'K')
2) The algorithm selects initial centre points

called centroids or mean values or medoids
3) Assign the objects or observations to the nearest centre (using diff dist measures)
4) update & repeat the clusters centres are recalculated based on the new members of observations are reassigned iteratively until the cluster centres stop changing or using an evaluation criterion is met (if within cluster var is minimized)

$m_1 = 3$ $m_2 = 11$

Data points	$d_1 = \|x - m_1\|$	$d_2 = \|x - m_2\|$	cluster
2	1	9	c_1
4	1	7	c_1
10	7	1	c_2
12	9	1	c_2
3	0	8	c_1
20	17	9	c_2
30	27	19	c_2
11	8	0	c_2
25	22	14	c_2

$C_1 = \{2, 4, 3\}$ $m_1 = \frac{2+4+3}{3} = 3$

$C_2 = \{10, 12, 20, 30, 11, 25\}$

$m_2 = \frac{10+12+20+30+11+25}{6} = 18$

Dtc DP	d(x-3) m_1	d|x-18| m_2	clus
2	1	16	c_1
4	1	14	c_1
10	7	8	c_1
12	9	6	c_2
3	0	15	c_1
20	17	2	c_2
30	27	12	c_2
11	8	7	c_2
25	22	7	c_2

$C_1 = \{2, 4, 10, 3\}$

$C_2 = \{12, 20, 30, 11, 25\}$

$m_1 = 4.75$

$m_2 = 19.$

2 Dte move

Data points	d_1	d_2	C
2	2.75	17.6	c_1
4	0.75	15.6	c_1
10	5.25	9.6	c_1
12	7.25	7.6	c_1
3	1.75	16.6	c_1
20	15.25	0.4	c_2
30	25.25	10.4	c_2
11	6.25	8.6	c_1
25	20.25	5.4	c_2

$$c_4 = \frac{2+4+10+12+3+11}{6}$$

$c_1 = 7$

$c_2 = 25$

Data point	d_1	d_2	C
2	5	23	C_1
4	3	21	C_2
10	3	15	C_1
12	5	13	C_1
3	4	22	C_1
20	18	5	C_2
30	23	5	C_2
11	4	14	C_1
25	18	0	C_2

$C_1 = \{2, 4, 10, 12, 3, 11\}$

$C_2 = \{20, 30, 25\}$

K means split n observation into exactly K clusters. Each point joins the cluster with the nearest centroid. K means ⇒ Spherical

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Density-based clustering: Finding clusters of any shape (arbitrary)

1) Density-based clustering is an unsupervised learning approach which groups data points into dense separated by sparser region treated as noise or outlies. This method is useful in identifying arbitrarily shaped clusters (S-shaped or uneven shaped clusters)

Applications:- Geospatial analysis, Anomaly Fraud detection- isolating credit card transaction, network traffic.

density based clustering operates on the premise that the cluster grows as long as the density (no of points or the no. of points within a certain neighbourhood exceeds a set threshold.

Most of the density based algorithms classify data points into 3 specific types based on local density radius

Neighbourhood ⇒ ϵ neighbourhood ⇒ neighbourhood of a point

Density around p ⇒ how many points inside the neighbourhood.

3 specific types ⇒ 1) core points: points that have atleast a minimum no. of neighbours (minpts) within a specific radius (ϵ)

2) Border points: points that fall within the ϵ radius of a core point but do not have enough neighbours themselves to qualify as core point.

3) Noise points (outliers): points that are neither core nor border points i.e., the points which are in low density regions or are explicitly excluded from the clusters.

There are 3 prominent density-based algorithms used in ML; each handling density variations differently.

1) Density based spatial clustering of applications with noise (DBSCAN)
This method uses a single fixed radius and threshold (MinPts) to discover clusters.

2) Hierarchical (HDBSCAN)
It is a self-adjusting version that converts DBSCAN into a hierarchical algorithm.

3) Ordering points to identify the clustering structure (OPTICS)
It is a multi scale approach that creates reachability plot mapping the distance b/w neighbouring features.

practical considerations → Scale the features, distance metric, high dimension, missing values.

Dimensionality reduction is the process reducing no. of features or dimensions in a dataset while retaining as much info as possible This can be done for a variety of reasons such as to reduce the complexity of the model to improve the performance a learning algorithm or to make it easier visualize the data. High dimensional data leads to overfitting where the model fits the training data too closely but does not generalize well to the new data. The curse of dimensionality is a common problem in ML when the performance of the model the deteriorates as the no of feature increases.

Dimensionality reduction is of two types:-

1) Feature Selection

Selecting the subset of relevant features in a data set to build a model of high accuracy (Selecting optimal feature from input dataset)

methods:-

1) Filter method → correlation
⇒ chi-square test
⇒ ANOVA
⇒ Info gain etc ---

2) Wrapped method:- ⇒ Forward selection
⇒ Backward Elimination
⇒ Step wise method Bidirectional elimination

3) Embedded method:- ⇒ Ridge regression
⇒ Lasso regression

2) Feature extraction:- It is the process of transforming this phase containing many dimensions into space with fewer dimension

methods:-

1) PCA (Principal component analysis)
2) Linear discriminant Analysis (LDA)
3) Kernel PCA
4) Quadratic discriminant Analysis (QDA)

Advantages of dimensionality reduction:-

1) Helps in data compression
2) Reduces computational time
3) Helps to remove redundant features
4) Improved visualization
5) Prevents overfitting of parameters
6) It can be used in preprocessing step

Disadvantages :-

1) Loss information

PCA:- It is the most popular feature extraction technique for unsupervised dimensionality reduction * PCA standardises the data.

* Identifies the dim where the data varies the most & project the data on to this new axis called principal components. The first Principal component captures the max possible variables in the dataset. The 2nd component captures the next highest variance & so on

Steps:- in PCA :-

S-1:- consider the data points

S-2: calculate the mean of the data points

S-3: calculate the covar matrix

S-4: Find the Eigenvalues & Eigen vectors

S-5: Find the principle components (uncorrelated)

(x_1, x_2) : $\overset{x}{(4,11)}$; $\overset{y}{(8,4)}$; $(13,5)$; $(7,14)$

$$\bar{x}_1 = \frac{4+8+13+7}{4} = 8 \qquad \bar{x}_2 = \frac{11+4+5+14}{4} = 8.5$$

$$S = \frac{1}{n-1} \Sigma (x_i - \bar{x})(y_i - \bar{y}) = \begin{pmatrix} 14 & -1 \\ -1 & 23 \end{pmatrix}$$

$|S - \lambda I| = 0$ (P) $\begin{vmatrix} 14-\lambda & -1 \\ -1 & 23-\lambda \end{vmatrix} = 0$

$\lambda^2 - 37\lambda + 321 = 0$

$\lambda = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = 30.3849, 6.6151$

$\lambda_1 = 30.3849 \quad \lambda_2 = 6.6151$

$\lambda = 30.3849$

Eigen vectors $E_1 = \begin{pmatrix} 0.5574 \\ 0.8303 \end{pmatrix}$

$\lambda_2 = 6.6151$

$E_2 = \begin{pmatrix} 0.8303 \\ 0.5574 \end{pmatrix}$

centred datapoint $= \begin{pmatrix} 4-8, 11-8.5 \\ 8-8, 4-8.5 \end{pmatrix} = \begin{pmatrix} -4, 2.5 \\ 0, -4.5 \end{pmatrix}$

projection $\begin{bmatrix} 0.5574 & 0.8303 \end{bmatrix} \begin{bmatrix} -4 \\ 2.5 \end{bmatrix} \approx 4.3052$

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Singular value decomposition:-(SVD)

SVD is a fundamental matrix factorisation technique in linear algebra. SVD reduces dimensionality in MS by factorizing a data matrix into lower rank approximations retaining only the most dominant patterns by discarding the noise. It can be viewed as a method for transforming correlated variables into a set of uncorrelated variables that vector expose the various relationships among the original data items it is an method for identifying ordering the dimen -sion along which data points exhibit the most variation. With SVD it is possible to find the best approximation of the original data points using fewer dimensions. Hence SVD is used for data reduction. SVD factorizes a mxn data matrix A into three matrices

$$A = U_{mxm} \Sigma_{mxn} V^T_{nxn}.$$

where, the columns of U are orthogonal eigenvectors of AA^T, the columns of V are orthogonal eigenvectors of A^TA & eigenva -lues $\lambda_1, \lambda_2 \ldots \lambda_r$ of AA^T. are $|AA^T - \lambda I| = 0$

& Find eigenvalue

(or) U is a left orthogonal matrix, Σ is a diagonal matrix of singular values, V^T is transpose of a right orthogonal matrix.

Steps:-

1) perform SVD i.e, factorise the original matrix A into $U \Sigma V^T$ (3 matrices)

2) Select top 'k' components i.e choose a reduced target dimension 'k' where $k \leq \min(m, n)$ based on cumulative variance or Screeplot (elbow points based on that we can choose)

3) Truncate the matrices that is keep only top 'k' singular values in Σ_k the first 'k' columns of U_k & first 'k' rows of V_k^T

4) project the data: compute the lower dimension represent B using $B = U_k \Sigma_k$

or $A_k = U_k \Sigma_k V_k$.

consider $A = \begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix}$ $A^T = \begin{bmatrix} 1 & 1 \\ 2 & 1 \end{bmatrix}$

① $A^T A = \begin{bmatrix} 1 & 1 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 3 \\ 3 & 5 \end{bmatrix}$

$A^T A = \begin{bmatrix} 1 & 1 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 3 \\ 3 & 5 \end{bmatrix}$

② Find the Eigen values.

$$|A^TA - \lambda I| = 0$$

$$\left|\begin{bmatrix}2 & 3\\ 3 & 5\end{bmatrix} - \lambda\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}\right| = 0 \quad \left|\begin{bmatrix}2 & 3\\ 3 & 5\end{bmatrix} - \begin{bmatrix}\lambda & 0\\ 0 & \lambda\end{bmatrix}\right| = 0$$

$$= \left|\begin{bmatrix}2-\lambda & 3\\ 3 & 5-\lambda\end{bmatrix} = 0\right|$$

$$(2-\lambda)(5-\lambda) - 9 = 0$$

$$10 - 2\lambda - 5\lambda + \lambda^2 - 9 = 0$$

$$\lambda^2 - 7\lambda + 1 = 0$$

$$= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \qquad a = 1 \quad b = -7 \quad c = 1$$

$$= \frac{7 \pm \sqrt{(-7)^2 - 4(1)(1)}}{2(1)}$$

$$= \frac{7 \pm \sqrt{49 - 4}}{2} = 7 \pm \frac{\sqrt{45}}{2}$$

$$= 7 \pm \frac{6.708}{2} \Rightarrow 7 \pm 6.686$$

$$= 6.686 \mid 0.145 \ \#$$

$\lambda_1 = 6.86$ $\qquad \sigma_1 = \sqrt{\lambda_1}$

$\lambda_2 = 0.145$ $\qquad \sigma_2 = \sqrt{\lambda_2}$

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Reinforcement learning

agent → A learner that make decisions

environment → everything the agent interacts with

States S, Action a, Reward r, policy π,

Value V(s) Q-value Q(s,a)

Reward types :-

Immediate ~~Feed~~ | Sparse
Delayed | Reward shaping

Bagging ⇒ Bootstrap (→ random samples) + Aggregating

Bagging - Independent model. Boosting - sequential model.
↳ Reduce variance ↳ Reduce bias

Boosting vocabulary :-

weak learner | Error correction
Strong learner | Observation weights
Sequential learning | Learning rate

Ada Boost - Adaptive Boosting

$e^{-(\beta_0 + \beta_1 x_1}$

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Activation functions in MLP's

1) Sigmoid: $\sigma(z) = \frac{1}{1+\exp(-z)}$

2) Tanh: $\tanh(z) = \frac{\exp(z) - \exp(-z)}{\exp(z) + \exp(-z)}$

ReLU (Rectified linear unit): ReLU(z) = max(0, z)

	ANN	CNN	RNN
Best for	Tabular data	Images	Sequences
Key idea	Layer of neurons	Sliding filters	Hidden-state memory
Memory	NO	NO	Yes.
Weight sharing	NO	Across space	Across time
Eg	Exam result	cat vs dog	next word.

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Formula of a neuron.

$$z = \sum_{i=1} w_i x_i + b;$$

Activation function

Sigmoid → 0 to 1. | tanh → -1 to 1

ReLU → pass the +ve, block the -ve

$f(z) = \max(0, z)$ ReLU

Soft max function: 0 : 1 sum = 1

convolution neural network → try to understand the pattern

Extracting from an image → Feature Extraction